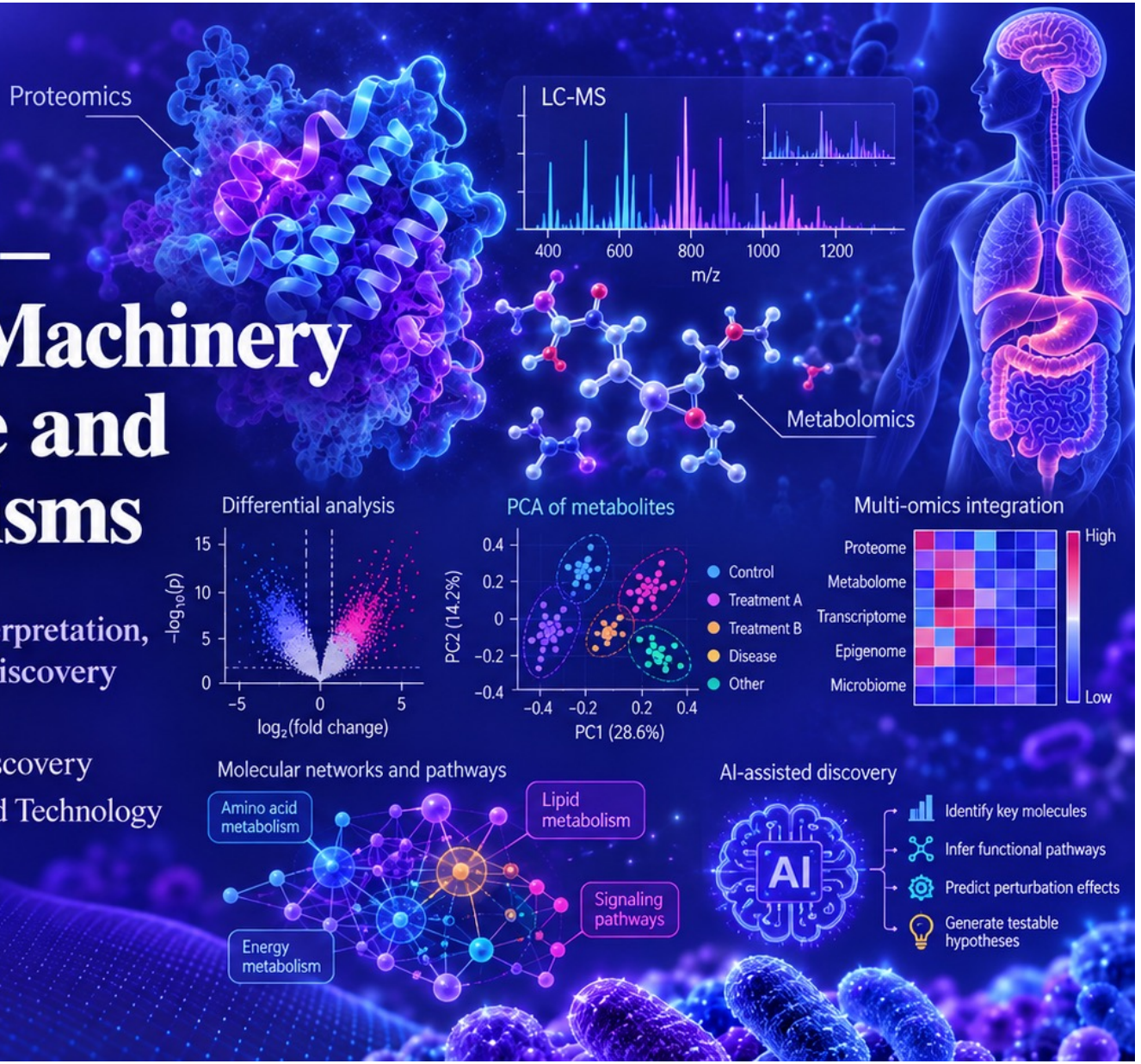
Week 7

Functional Omics – From Molecular Machinery to Chemical State and Testable Mechanisms

Proteomics, Metabolomics, Quantitative Interpretation, Multi-Omics Integration, and AI-Assisted Discovery

Bioinformatics: From Multi-Omics Data to Discovery
Dr. Liwei Xie | Shenzhen University of Advanced Technology

Two 40-Minute Sessions | Fall 2026



Functional Omics: Proteomics & Metabolomics

From measurement to mechanism, phenotype, and discovery

1 DNA



- **potential**

genetic information

2 RNA



- **program**

transcriptional output

3 Protein



- **machinery**

functional molecules

4 Metabolite



- **chemical state**

small molecules and pathways

5 Phenotype



- **outcome**

observable traits and functions

Week 7 focus

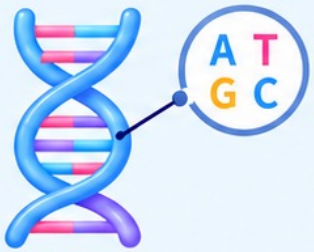
- Which molecular machinery is operating?
- What chemical state has emerged?

This week moves from molecular potential to functional state.

Why do we need proteomics and metabolomics?

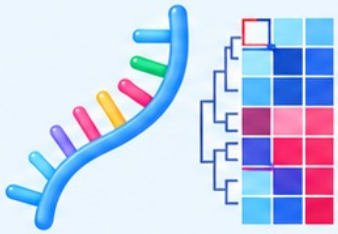
Earlier omics reveal potential and programs, but not yet the full functional state.

1 Genomics



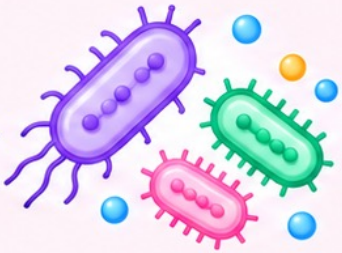
- **Tells us:** sequence and regulatory potential
- **Still missing:** is signaling actually changed?

2 Transcriptomics



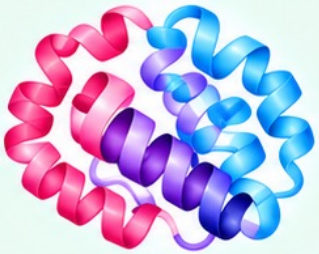
- **Tells us:** expression programs
- **Still missing:** is the pathway activated?

3 Microbiome



- **Tells us:** community structure and functional capacity
- **Still missing:** is the metabolite really produced?

4 Proteomics



- **Which molecular machinery is operating?**

enzymes, receptors, complexes, PTMs

5 Metabolomics



- **What chemical state has emerged?**

metabolites, pathways, host-microbe signals

Functional omics connects previous omics layers to mechanism and phenotype.

One Biological System, Five Information Layers

The same phenotype can be viewed through different omics lenses.

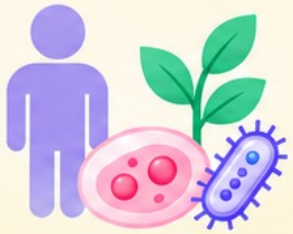
1 DNA
- variation / regulation



2 RNA
- expression program



Phenotype

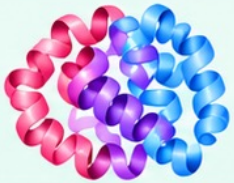


e.g., insulin resistance

3 Microbiome
- community and functional capacity



4 Protein
- molecular machinery



5 Metabolite
- chemical output



Upstream information

DNA and RNA capture genetic variation and expression programs.

Functional execution

Microbiome and protein layers reflect molecular and cellular machinery that execute biological functions.

Closest to phenotype

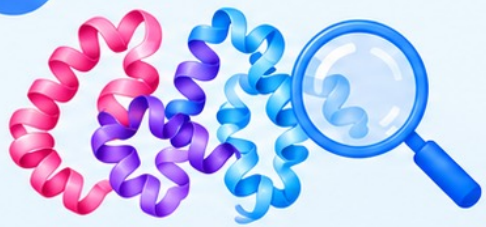
Metabolites are often the closest layer to the observable phenotype, integrating multiple upstream signals.

No single omics layer is the whole truth; each measures a different biological question.

The first decision: untargeted or targeted?

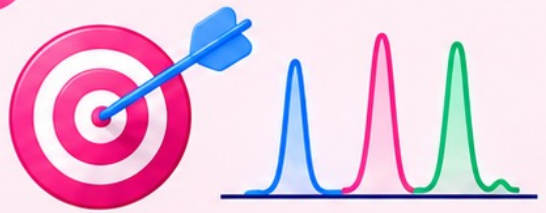
Choose breadth or precision according to the biological question.

1 Untargeted proteomics



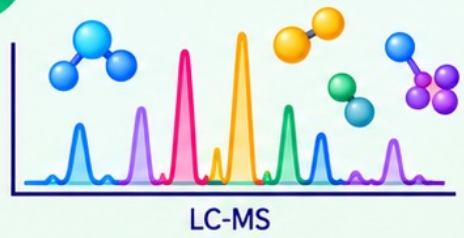
- broad coverage
- hypothesis generation
- relative quantification

2 Targeted proteomics



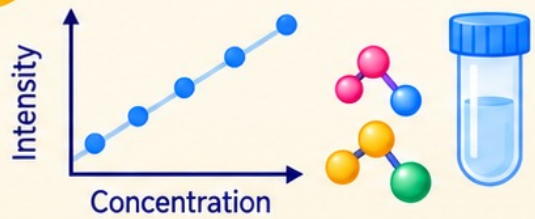
- PRM / SRM / MRM
- high sensitivity
- candidate validation

3 Untargeted metabolomics



- many features
- discovery mode
- annotation burden

4 Targeted metabolomics



- defined analytes
- stronger quantification
- translational robustness

Discovery / Breadth → **Validation / Precision**

Typical workflow: untargeted discovery → targeted validation.

The assay should follow the question: **discover first, validate second** when needed.

Targeted vs. Untargeted Proteomics and Metabolomics

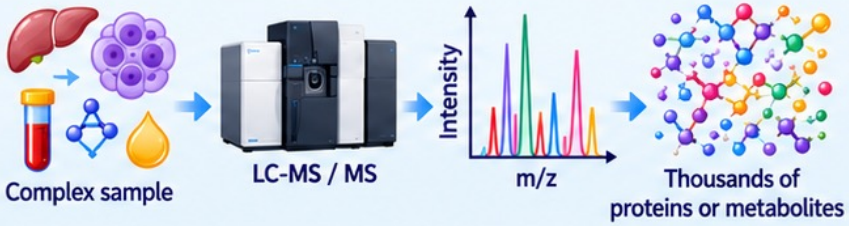
Different strategies answer different biological questions.

How to read this slide

- **Untargeted** = discovery / broad coverage.
- **Targeted** = validation / sensitive quantification.
- Choose the assay by the biological question.

1 Untargeted (Discovery)

Global profiling to capture as many molecules as possible.



2 Targeted (Hypothesis-driven)

Focused quantification of predefined molecules.



	Untargeted (Discovery)	Targeted (Hypothesis-driven)
Goal	Broad, unbiased profiling to discover novel molecules and pathways.	Accurate and sensitive quantification of predefined molecules.
Coverage	Global coverage (thousands of proteins/metabolites).	Limited to predefined targets (tens to hundreds).
Quantification	Relative or semi-quantitative (label-free or isotope-labeled).	Highly sensitive and precise (e.g., MRM/PRM/SRM, isotopic standards).
Strengths	Hypothesis generation Discover novel biomarkers and pathways Capture unexpected changes and network-level information	High sensitivity and quantitative accuracy Reproducible and robust Ideal for validation and clinical translation
Limitations	Lower sensitivity for low-abundance molecules Higher missing values and batch effects More complex data analysis and annotation	Limited to predefined targets Cannot discover new molecules Requires prior knowledge and standardized assays
Typical applications	Biomarker discovery Pathway and network analysis Mechanism exploration and multi-omics integration	Biomarker validation Targeted pathway analysis Clinical and large-scale Pharmacokinetic and mechanistic studies

When to choose?

Choose untargeted if:

- You want to explore unknown biology.
- You need comprehensive coverage.
- You are at the discovery or hypothesis-generating stage.

Choose targeted if:

- You have predefined candidates.
- You need high sensitivity and precise quantification.
- You are validating biomarkers or translating to clinical studies.

A combined strategy is often optimal:

- Untargeted for discovery.
- Targeted for validation.
- Integration provides the most robust biological insights.

What does a mass spectrometer actually measure?

In bottom-up proteomics, protein identity is inferred from peptide ions and spectra.

1 Protein sample



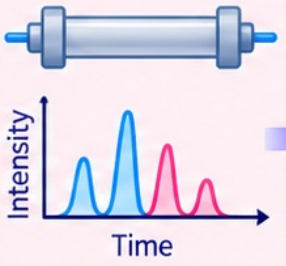
- tissue, cells, plasma, etc.
- complex protein mixture

2 Proteolytic digestion



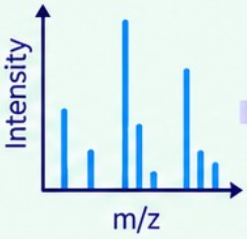
- trypsin (e.g.) cleaves proteins into peptides
- mixture of peptides

3 LC separation



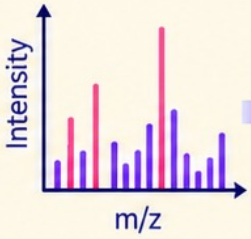
- liquid chromatography separates peptides in time
- peptides elute one by one (or in groups)

4 MS1



- mass-to-charge of peptide ions

5 MS2



- fragment ions provide sequence evidence

6 Peptide identification



- match MS/MS spectra to a peptide database
- identify peptide sequences

7 Protein inference



- peptides are mapped back to proteins
- infer protein identity and abundance

MS does not directly see an intact protein in routine bottom-up proteomics.
The instrument measures peptide ions and their fragments, and protein identity is inferred computationally.

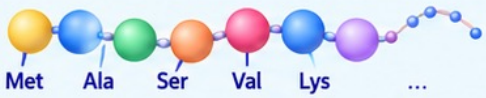
Proteomics is an inference chain: ions → peptides → proteins.

A protein is more than a sequence

Sequence, structure, and conformation determine what we can detect in proteomics.

1 1D: Primary structure

A linear sequence of amino acids.



M – A – S – V – K – ... (amino acid sequence)

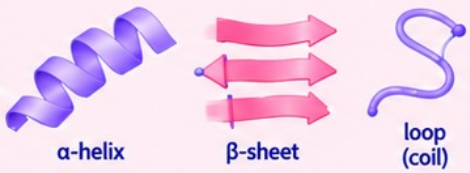
- Defines the protein's identity
- Determines theoretical peptide sequences for MS detection
- Substrate for enzymatic digestion (e.g., trypsin cleaves after K/R)
- Used for protein inference and database searching

Key considerations for proteomics

- Sequence determines which peptides can be generated and detected.
- Variants (SNPs, isoforms) change peptide sequences.
- Sequence alone does not reveal structure, modifications, or function.

2 2D: Secondary structure

Local folding patterns stabilized by backbone hydrogen bonds.



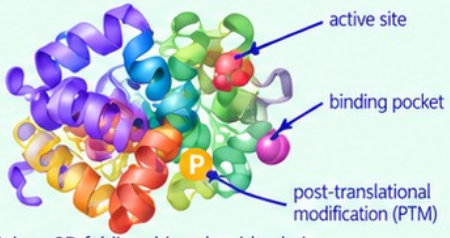
- Regular patterns formed by hydrogen bonds between backbone atoms
- Common motifs: α-helix, β-sheet, loop
- Influences protease accessibility and peptide generation
- May affect ionization efficiency and fragmentation behavior

Key considerations for proteomics

- Structured regions may be less accessible to digestion.
- Flexible/loop regions are often better detected.
- Secondary structure can influence peptide detectability and reproducibility.

3 3D: Tertiary structure

The full 3D conformation of one polypeptide chain.



- Unique 3D folding driven by side-chain interactions (hydrophobic, ionic, disulfide bonds, etc.)
- Creates functional sites (e.g., binding pockets, active sites)
- Determines accessibility to enzymes and antibodies
- Hosts post-translational modifications (PTMs)
- Can exist in multiple conformations (conformational dynamics)

Key considerations for proteomics

- 3D structure affects digestion efficiency and coverage.
- PTMs often occur on surface or flexible regions.
- Conformation can change with conditions, bringing environment.
- Native structure information requires specialized approaches (e.g., cross-linking MS, HDX-MS).

Why structure matters in proteomics

Detection & coverage

- Structured or buried regions may be poorly digested, leading to lower coverage.
- Flexible regions are usually detected more easily.

PTM localization

- PTMs depend on the 3D context (surface exposure, local environment).
- Structural information helps interpret PTM function and cross-talk.

Protein interactions

- Protein–protein interactions depend on 3D interfaces.
- Cross-linking MS and structural proteomics provide complementary information.

Function and phenotype

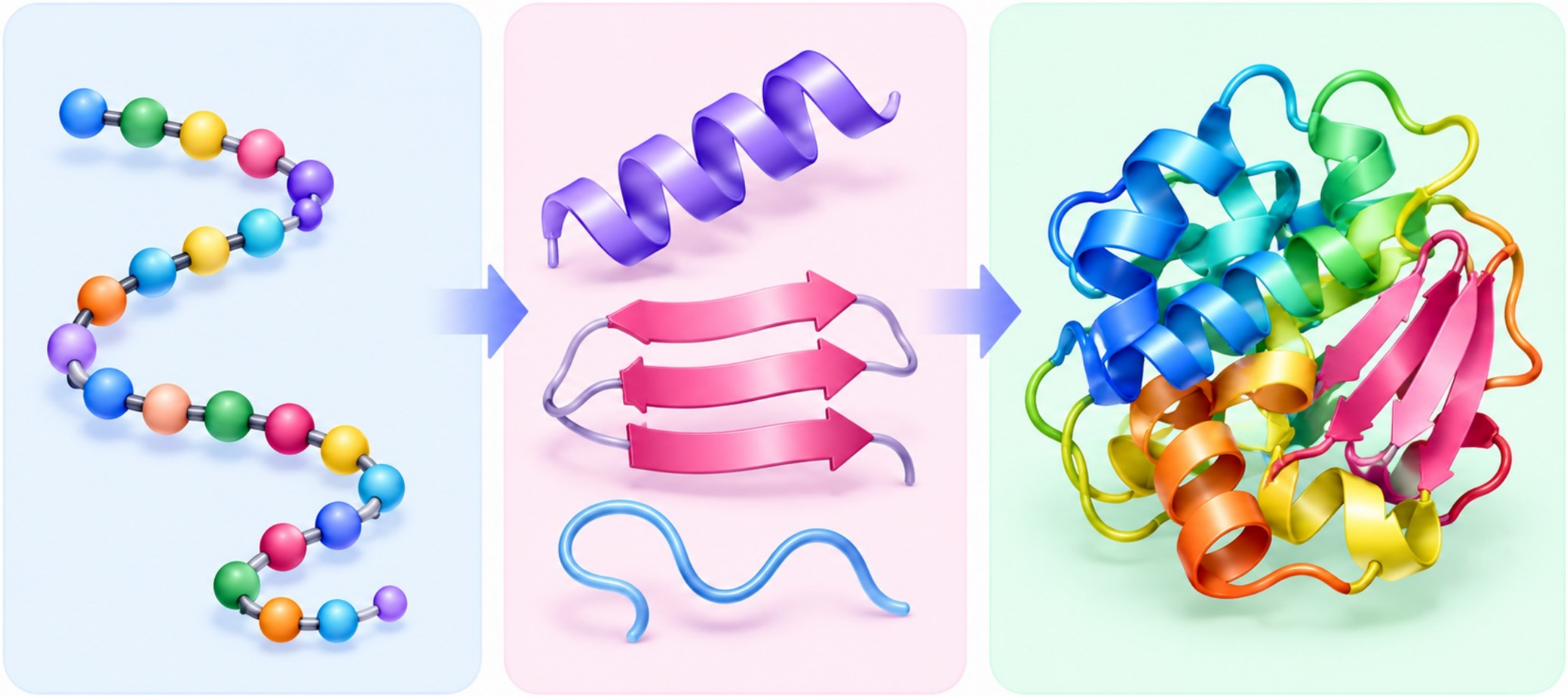
- Structure determines activity, stability, and interaction networks.
- Structural changes (e.g., mutation, PTM, ligand binding) can alter function and disease relevance.

Key takeaways for proteomics experiments

1. Sequence (1D) tells us what peptides we can detect.
2. Structure (2D/3D) affects which peptides are actually detected.
3. PTMs and interactions depend on 3D context.
4. Interpretation of proteomics data requires considering sequence, structure, and biology together.

In proteomics, the same protein can look very different depending on how it is prepared, detected, and interpreted.

From one to three dimentional structure of a protein



From a sample to a protein matrix

Experimental preparation shapes what the proteomics result can mean.

1 Sample



- tissue (biopsy)
- cells (culture)
- plasma / serum
- microbiome sample (stool)

2 Lysis & extraction



- cell lysis
- protein extraction
- detergents / buffers
- protein solubilization

interaction loss
protein complexes may disassemble

3 Reduction / alkylation



- reduce disulfide bonds (DTT/TCEP)
- alkylate cysteines (iodoacetamide)
- stabilize peptides

incomplete reaction
variable modification patterns

4 Trypsin digestion



- digest proteins to peptides
- typically trypsin (K/R cleavage)
- generate peptide mixture

missed cleavage
incomplete digestion alters coverage

5 Peptide cleanup + LC-MS/MS



- peptide cleanup (desalting / fractionation)
- liquid chromatography (LC) separation
- tandem mass spectrometry (MS/MS)

batch drift
instrument and run variability

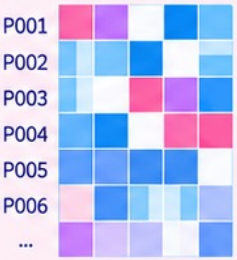
6 Identification & quantification

Protein	Sample 1	Sample 2	Sample 3
P001	12.5	8.1	0.0
P002	5.2	3.9	7.3
P003	0.0	1.1	2.6
P004	8.4	9.2	6.1
...	...	...	...

- match spectra to peptides / proteins
- estimate abundances (intensity or counts)
- filtering and normalization

missingness
not all proteins detected in all samples

7 Protein matrix



- proteins (rows)
- samples (columns)
- abundance values
- input for downstream analysis

Why preparation matters

- **sample type differs**
  tissues, cells, plasma, and microbiome samples have different protein content and properties.
- **protocol introduces bias**
  each step (lysis, digestion, cleanup, MS) can favor or lose certain proteins.
- **QC comes before biology**
  assess sample and instrument quality before interpreting biological results.

A protein matrix is the endpoint of many experimental choices, not a raw truth.

Do not mix the concepts

Acquisition, quantification, and targeting are different decisions in proteomics.

1 Acquisition

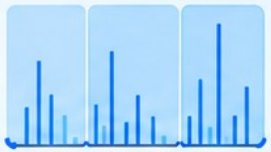
How to select and fragment ions from the sample.

DDA



- Data-dependent acquisition
- Selects the most abundant precursors for fragmentation.

DIA



- Data-independent acquisition
- Fragments all ions within predefined m/z windows.

2 Quantification

How to compare peptide or protein abundance across samples.

LFQ



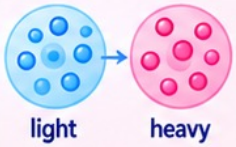
- Label-free quantification
- Compares signal intensities across runs.

TMT



- Isobaric tags for multiplexing
- Compare multiple samples in one MS run.

SILAC

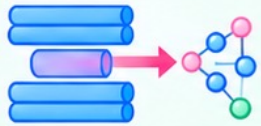


- Metabolic labeling with stable isotopes
- Compare samples mixed before MS analysis.

3 Targeted validation

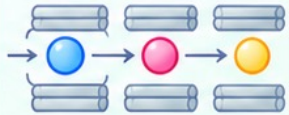
How to specifically measure predefined peptides or proteins.

PRM



- Parallel reaction monitoring
- High-resolution targeted MS/MS of predefined peptides.

SRM/MRM



- Selected reaction monitoring (multiple)
- Targeted MS/MS using predefined precursor → fragment transitions.

Ask three different questions:

1. Which ions are fragmented?
2. How are samples compared?
3. Are analytes predefined?

DDA, DIA, LFQ, TMT, and PRM describe different dimensions of the workflow, not interchangeable labels.

Gene does not equal one protein object

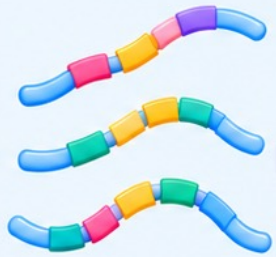
Proteoforms and context help explain why RNA abundance does not fully predict function.

1 Gene



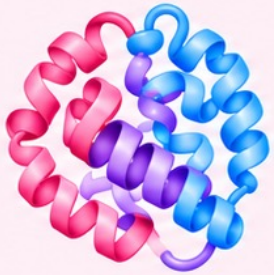
- Genetic information
- One gene can give rise to multiple products

2 RNA



- Transcripts (splice isoforms)
- Different isoforms can produce different proteins

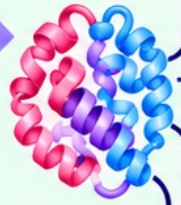
3 Protein abundance



- Amount of protein produced
- Measurable by proteomics (e.g., MS)

4 Proteoforms

A gene can produce many proteoforms with different properties.



Post-translational modifications (PTMs)
e.g., phosphorylation, acetylation, glycosylation

Complex membership
e.g., protein complexes, interacting partners

Localization
e.g., cytosol, nucleus, membrane, organelles

Turnover / stability
e.g., half-life, degradation rate

RNA unchanged? Biology may still change.

- Same RNA levels can yield different proteoforms (e.g., via PTMs, localization or complex formation).
- Protein state and context can alter function without changes in RNA abundance.

mRNA–protein correlation is informative but incomplete.

- RNA levels provide important context.
- Protein state and proteoforms often explain functional differences.

Function depends on proteoforms and protein state, not only abundance.

Proteomics is not one experiment

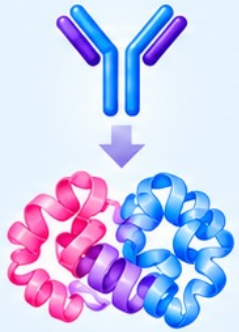
Choose the assay from the scientific question.

1 How much?



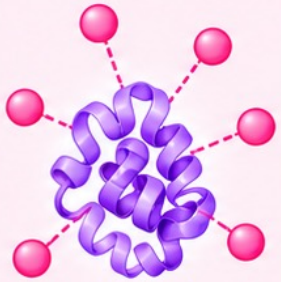
- Total proteomics

2 Who binds?



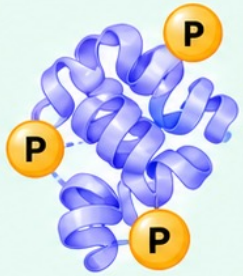
- Co-IP / AP-MS

3 Who is nearby?



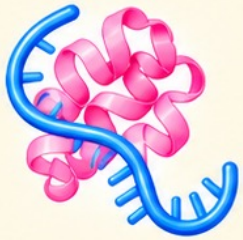
- BioID / TurboID / APEX2

4 Which signaling site changes?



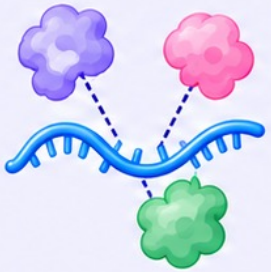
- PTM proteomics

5 Which RNAs bind this protein?



- RIP / CLIP

6 Which proteins bind this RNA?



- RNA pull-down-MS

7 Are variant peptides presented?



- Proteogenomics / immunopeptidomics

A mechanism study often combines several of these assays.

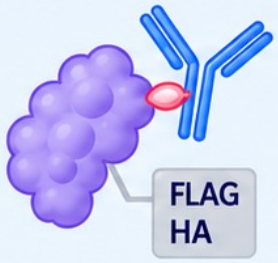
Start from the **biological question**, then select the **proteomics strategy.**

Co-IP and AP-MS: same complex does not mean direct binding

Enrichment supports biochemical co-association under the tested conditions.

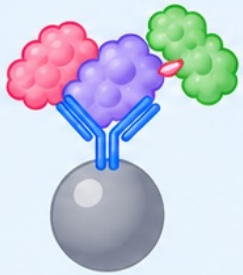
Experimental logic

1 Bait protein
antibody or tag



- endogenous antibody or epitope tag
- immobilize on beads

2 Enrichment
with matched controls



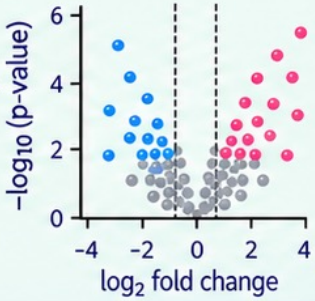
- capture bait and interactors
- parallel controls
- wash to reduce non-specific binding

3 LC-MS/MS
identify and quantify proteins



- digest proteins
- LC-MS/MS analysis
- peptide/protein identification and quantification

4 Enrichment comparison
bait vs controls



- enriched with bait
- enriched in control
- not significant

Interpretation

What it supports

- co-association under assay conditions
- candidate complex members
- reproducible enrichment vs controls

What it does NOT prove

- direct physical contact
- directionality
- functional relevance

Controls

- isotype / beads-only
- bait-null or untagged control
- input and unbound fractions

Co-IP/AP-MS suggests complex association; orthogonal validation is required for mechanism.

Co-IP vs proximity labeling

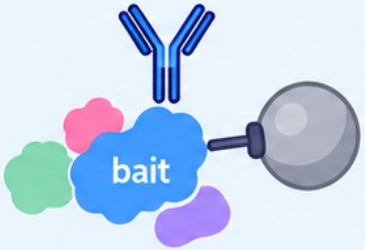
Same complex and same neighborhood are related ideas, but they are not the same biological claim.

1 Scientific question



- who is with my bait?
- under native conditions

2 Co-IP / AP-MS



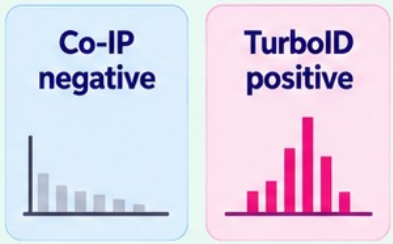
- captures co-associated proteins
- requires preserved complexes
- good for stable associations

3 BioID / TurboID / APEX2



- labels nearby proteins
- captures local neighborhood
- good for weak or transient proximity

4 A result pattern



Contradiction?

not necessarily

5 How to interpret

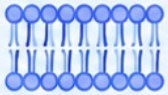


- direct binding is still unproven
- topology and labeling window matter
- use orthogonal validation

Possible explanation



transient contact



membrane proximity



weak interaction



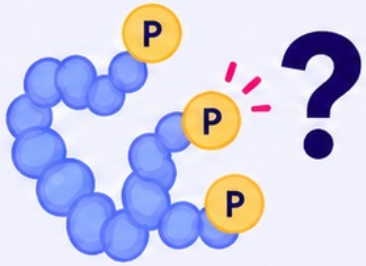
lysis-sensitive complex

Choose the assay by the biological question: **co-association and proximity are complementary, not interchangeable.**

PTM proteomics: abundance does not equal activity

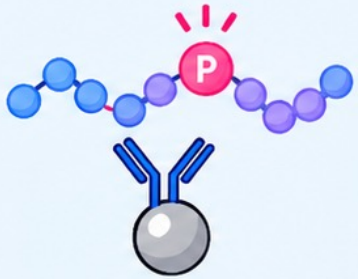
Site-specific modification data are informative only when interpreted with protein abundance, localization confidence, and biological context.

1 Question



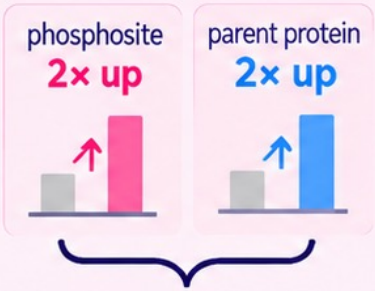
- which modification site changes?
- is signaling activated?

2 What is measured



- enriched modified peptides
- site-level intensities
- LC-MS/MS evidence

3 A common trap



activation?

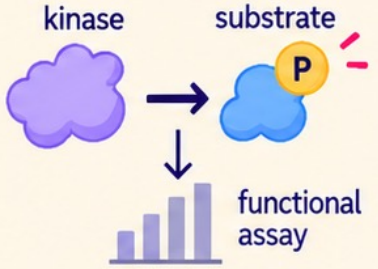
not automatically

4 What to record



- site localization confidence
- site multiplicity
- parent-protein abundance
- time course / perturbation

5 How to strengthen



- integrate total proteomics
- use kinase/substrate knowledge
- verify with phenotype or assay

Interpretation logic

Increasing evidence for biological activity.



signal only
site change alone

site + protein
site and total protein both change

site + pathway
consistent with pathway context

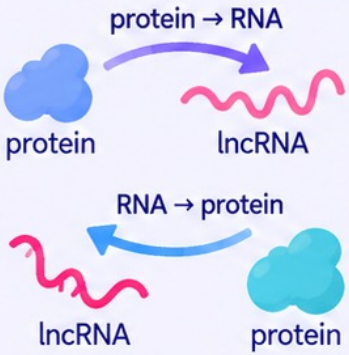
site + function
linked to phenotype or functional readout

A modified site is strongest when site dynamics, total protein, pathway context, and functional response agree.

ncRNA meets proteomics

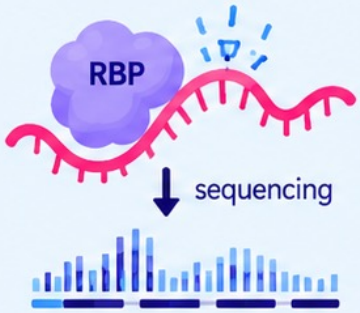
RNA–protein studies become mechanistic when the direction of the question and the downstream evidence are explicit.

1 Two directions



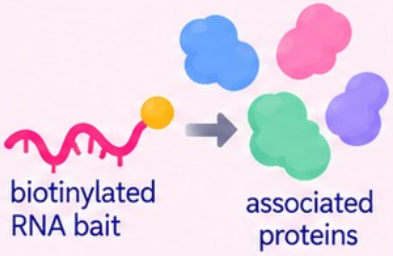
- ask the direction first
- different assays answer different questions

2 RIP / CLIP



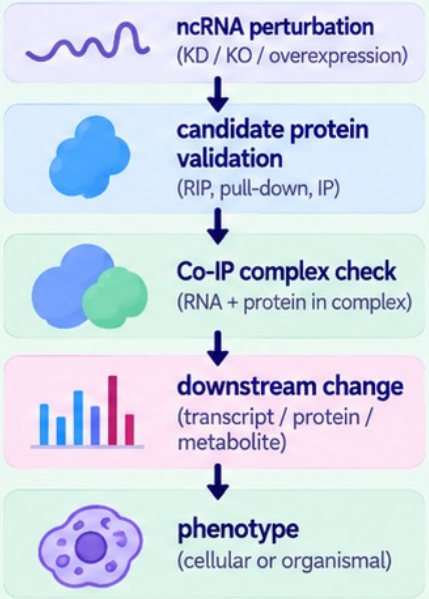
- selected protein → associated RNAs
- binding regions with CLIP variants
- not a global proteomics assay

3 RNA pull-down–MS



- selected RNA → associated proteins
- requires matched controls
- RNA structure and localization matter

4 Mechanism workflow



5 How to interpret



- association is not causality
- combine orthogonal evidence
- RNA abundance alone may not predict protein output

Examples of integrated mechanisms


lncRNA scaffold


miRNA target repression


circRNA protein/miRNA interaction

Start with the **direction of the biological question,** then link RNA–protein evidence to **downstream protein function and phenotype.**

Proteogenomics and immunopeptidomics

Prediction, expression, peptide evidence, presentation, and immunogenicity are different levels of evidence.

1 DNA mutation / noncanonical sequence



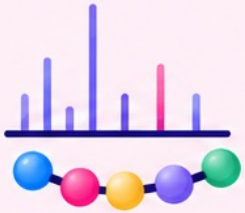
- somatic or germline variant
- noncanonical ORF (e.g., neoORF, fusion)
- expanded search space

2 RNA expressed



- transcript detected (RNA-seq)
- supports expression of mutated/variant sequence
- suggests potential translation

3 MS-supported peptide



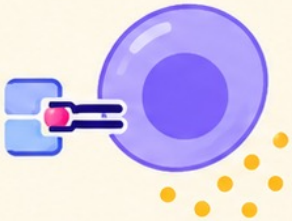
- peptide sequence identified by MS
- matches variant/ noncanonical sequence
- supported by spectra and quality metrics

4 HLA-presented peptide



- peptide detected on HLA (e.g., IP-MS)
- evidence of processing and presentation
- confirms cellular presentation

5 T-cell functional antigen



- T-cell recognition (e.g., activation, IFN-γ)
- demonstrates immunogenicity
- potential for immune targeting

Why this matters

- reduces the gap between predicted and observed antigens
- connects genome, transcriptome, and peptide evidence

Key interpretation rules

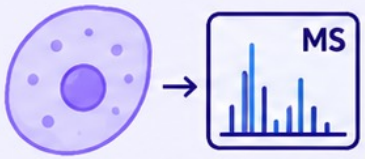
- larger search space raises false-assignment risk
- use decoy / entrapment strategies
- check peptide uniqueness and spectra
- presentation is not immunogenicity

Prediction is the start; confident claims require **matched multi-omics evidence and orthogonal validation.**

Frontier proteomics

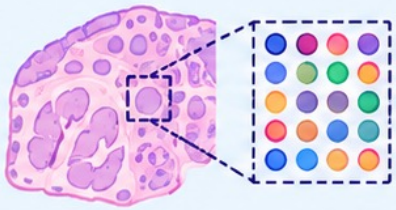
Single-cell, spatial, and structure-guided approaches extend proteomics from bulk averages to cell state, tissue context, and molecular design.

1 Single-cell proteomics



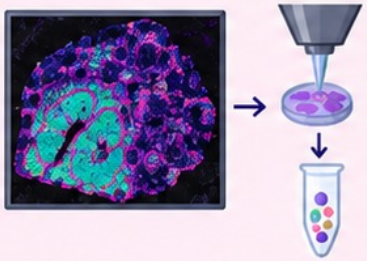
- very low sample amount
- sensitivity and sample-loss are central
- rare-cell phenotypes need validation

2 Spatial proteomics



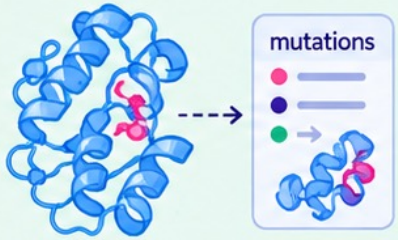
- where in the tissue?
- cell neighborhoods and morphology

3 Deep / imaging spatial methods



- multiplex antibody imaging
- LC-MS on selected regions

4 Structure-guided proteomics



- AlphaFold-type models for hypothesis generation
- design before DBTL

5 What to remember



- performance is contextual
- methods are complementary
- validation still matters

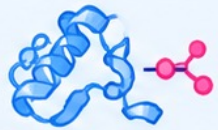
Three frontier questions



Which cell?



Where?



Which molecular design?

Frontier methods add **resolution and context**, but biological interpretation still depends on **careful controls and validation.**

Before biology: can you trust the data?

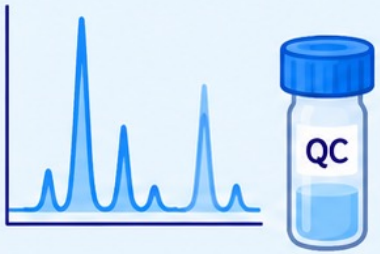
Quality control is the gate before interpretation in proteomics and metabolomics.

Proteomics QC



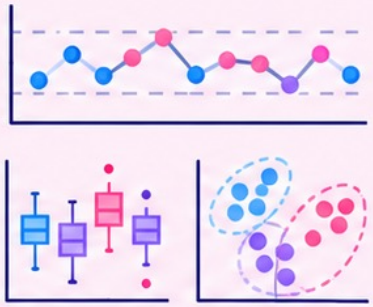
- IDs / FDR
- missingness
- batch and run drift

Metabolomics QC



- blanks
- pooled QCs
- internal standards
- RT drift

What to inspect



- signal stability
- run order effects
- sample outliers

What not to do



- jump straight to volcano plots
- ignore failed controls
- mix correction with biology

Decision



- repair and document
- exclude unstable signals
- only then test biology

QC mindset

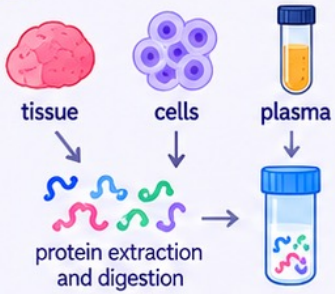
measure | inspect | decide | document

If QC is weak, the biological story is weak.

Proteomics QC: can you trust the protein matrix?

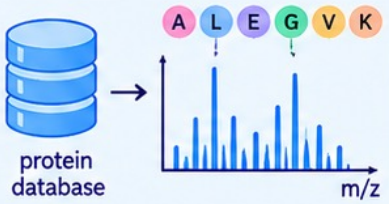
Quality control comes before differential abundance, pathway analysis, and biomarker claims.

1 Sample & prep QC



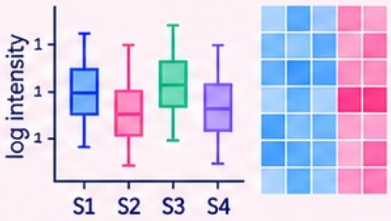
- sample identity / metadata
- protein yield and integrity
- lysis / digestion consistency
- labeling efficiency if TMT

2 Identification QC



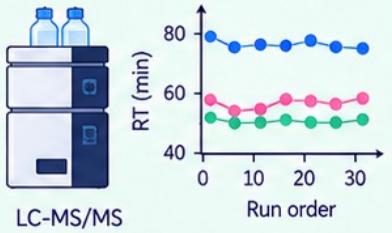
- PSM / peptide / protein IDs
- 1% FDR control
- missed cleavages
- contaminants and decoys

3 Quantification QC



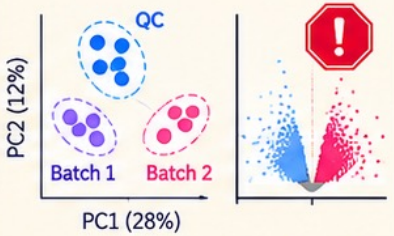
- intensity distribution
- replicate CV
- missing values
- normalization check

4 Run stability



- retention-time stability
- instrument drift
- batch / run-order effect
- internal standard or reference sample consistency

5 What to inspect before biology



- sample clustering vs batch
- outliers
- bridge channels / pooled refs
- do not jump straight to volcano plots

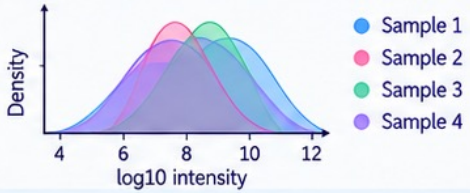
Typical proteomics QC readouts

IDs and FDR

PSMs	125,432
Peptides	18,927
Proteins	4,892
FDR (PSM)	0.8%
FDR (Protein)	0.7%

≤ 1% FDR

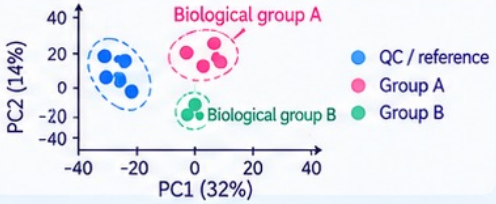
Intensity distribution across samples



Missingness heatmap



PCA: technical replicates cluster tightly



Common decisions

Repair & document
Address sample mix-ups, failed runs, instrument issues, and metadata problems.

Filter unstable proteins/peptides
Remove contaminants, high-missingness features, and poorly reproducible signals.

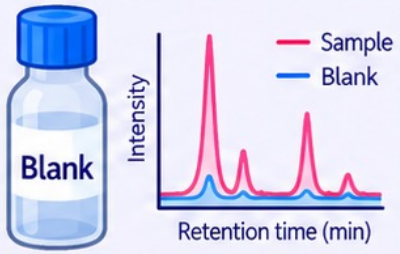
Only then test biology
Proceed to differential abundance, pathway analysis, and biomarker claims.

A protein matrix is an inference product; **weak QC weakens every biological claim.**

Metabolomics QC: can you trust the feature table?

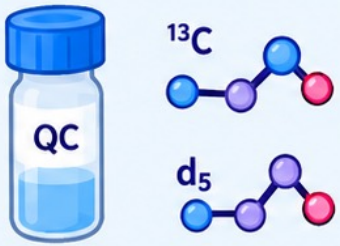
Before interpreting metabolites, verify chemistry, stability, and batch behavior.

1 Blank & contamination check



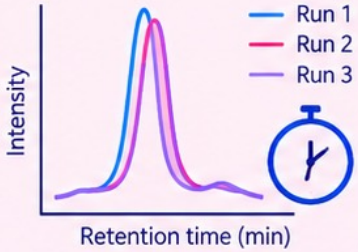
- blank subtraction
- background contaminants
- carryover
- solvent / column artifacts

2 Pooled QC & internal standards



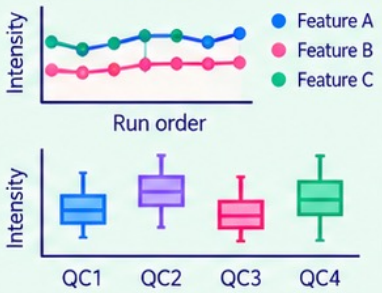
- pooled QC clustering
- internal standard response
- extraction consistency
- injection stability

3 Peak & alignment QC



- peak shape
- retention-time drift
- feature alignment
- adduct / isotope grouping

4 Quantitative stability



- QC CV or RSD
- run-order drift
- signal correction
- missing values

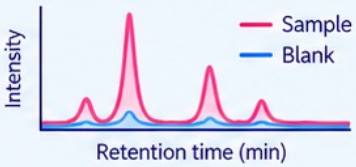
5 Annotation confidence



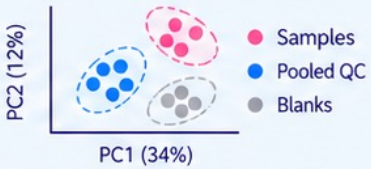
- mass is not identity
- MS2 / standard evidence
- isomer caution
- document confidence level

Typical metabolomics QC readouts

1. Blank vs sample peaks



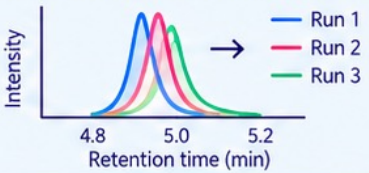
2. Pooled QC clustering (PCA)



3. Internal standard trend



4. RT drift / feature alignment



Common decisions

- remove contaminated / unstable features
- correct drift and document
- only then interpret pathways or biomarkers

A metabolite feature table mixes chemistry and biology; strong QC is the gate to **credible interpretation**.

A feature is not yet a metabolite

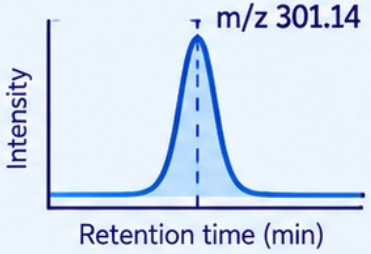
Untargeted metabolomics begins with signals; chemical identity requires graded evidence.

1 Sampling



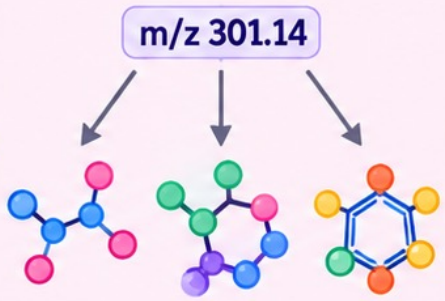
- matrix and extraction matter

2 LC-MS feature



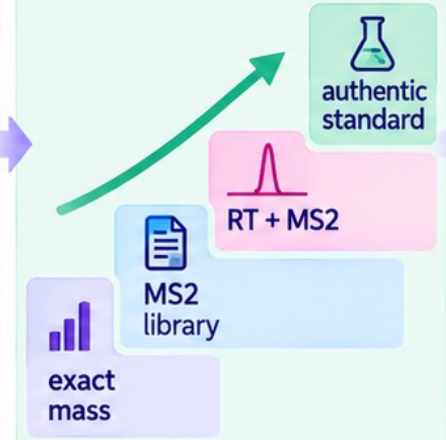
- retention time
- peak area
- adducts and isotopes

3 Why identity is hard



- multiple formulas or adducts
- isomers can share mass

4 Evidence ladder



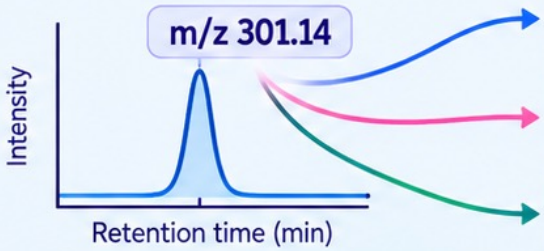
5 Interpretation

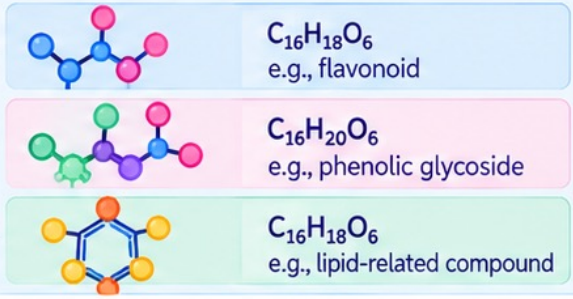


- report confidence level
- AI can prioritize, not confirm

Teaching example

One LC-MS feature can match multiple candidate structures.





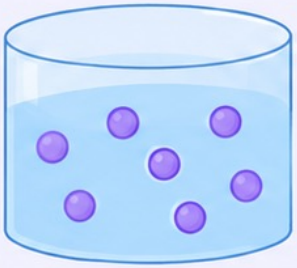
same measurement, different candidates

An LC-MS feature becomes a metabolite only when the evidence supports the name.

Concentration is not flux

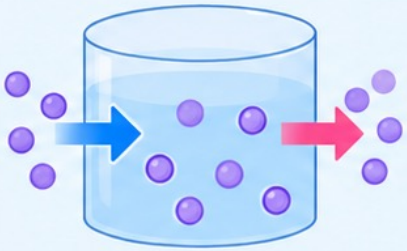
Pool size, labeling, and pathway rate are related but not interchangeable ideas.

1 Pool size



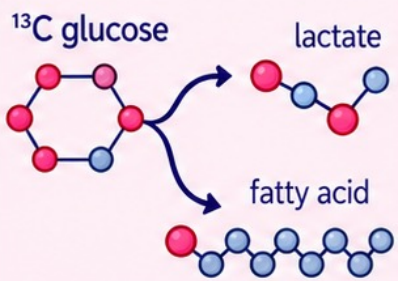
- how much is present?

2 Input vs output



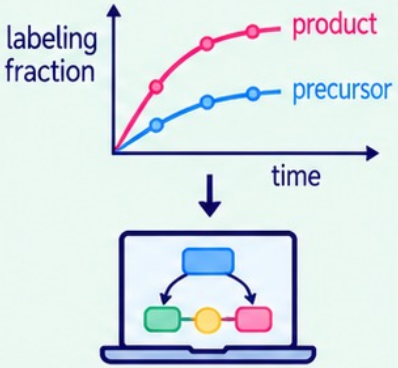
- higher abundance can reflect higher production
- or lower consumption

3 Stable-isotope tracing



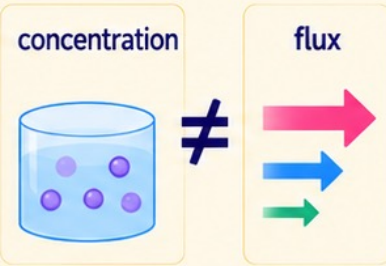
- tracks atom origin
- labeling fraction is not flux

4 Flux inference



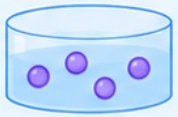
- needs time points
- needs precursor enrichment
- depends on assumptions

5 Interpretation



- abundance ≠ rate
- labeling ≠ final mechanism

Three different questions



How much?



Where did it come from?



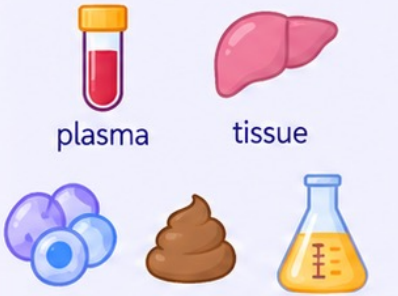
How fast is it moving?

Measure abundance, trace origin, and infer flux as **three distinct analytical layers.**

Sample preparation defines what you can see

In metabolomics, sampling, quenching, extraction, and storage shape the measurable chemical state.

1 Biological source



- different matrices
- different chemistry

2 Timing matters



e.g. after sampling:



- metabolism continues after sampling
- delay changes chemistry

3 Quench / extract



- stop metabolism fast
- choose solvent by analyte class

4 Storage and stability



- freeze-thaw affects signals
- oxidation / degradation risk

5 Interpretation



- sampling is part of the experiment
- technical variability can masquerade as biology

A practical rule



collect
standardized method

quench
stop metabolism

extract
appropriate solvent

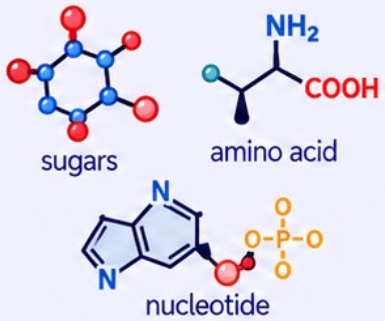
store
low temperature

The metabolome is highly sensitive to sample handling; **preparation is part of the biological measurement.**

Metabolite classes and what they mean for sample preparation

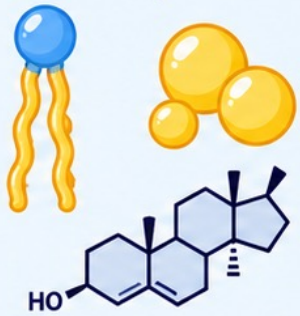
Polarity, stability, volatility, and abundance determine extraction strategy and analytical platform.

1 Polar metabolites



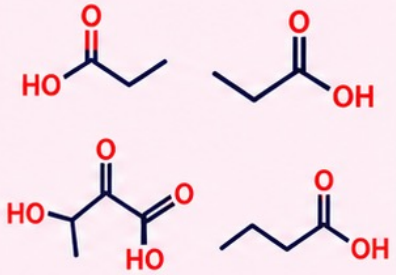
- amino acids, sugars, nucleotides
- highly hydrophilic
- fast turnover
- often cold aqueous-organic extraction

2 Lipids



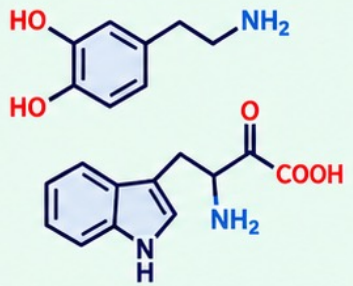
- phospholipids, TAGs, sterols
- hydrophobic
- often organic or biphasic extraction
- commonly RP-LC-MS

3 Organic acids / SCFAs



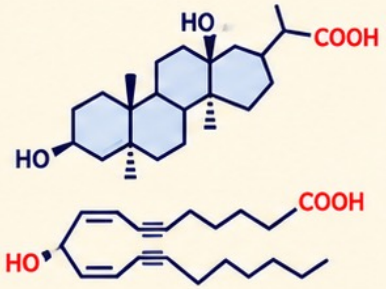
- TCA intermediates, lactate, SCFAs
- small and often very polar
- some are volatile or unstable
- may require derivatization or targeted assays

4 Biogenic amines & indoles



- amines, tryptophan metabolites
- often low abundance
- oxidation-sensitive
- acidified solvent + internal standards help

5 Bile acids & signaling metabolites



- bile acids, eicosanoids, other signals
- isomers and conjugates matter
- chromatographic separation is critical
- authentic standards strengthen identification

How chemistry guides preparation

Polarity
aqueous vs organic solvent choice

Stability
quench fast; keep cold; limit oxidation

Volatility
SCFAs and small acids may need special handling

Isomerism
same mass does not mean same metabolite

classify chemistry → choose solvent → choose platform → qualify confidence

In metabolomics, **chemical class is not decoration;** it determines what you can recover and what you can claim.

Biological sources require different metabolomics preparation

Matrix composition, biomass, and storage conditions shape quenching, extraction, cleanup, and normalization.

1 Plasma / serum



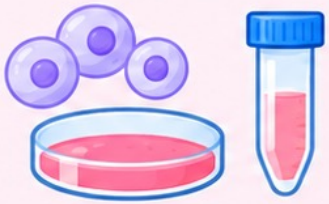
- protein-rich and salt-containing
- cold methanol or acetonitrile precipitation
- good for amino acids, lipids, bile acids
- watch hemolysis and batch handling

2 Tissue



- high enzymatic activity
- snap-freeze and keep cold
- rapid homogenization
- normalize by wet weight or protein

3 Cultured cells



- quench metabolism immediately
- wash carefully to avoid medium carryover
- extract with cold solvent
- normalize by cell number, protein, or DNA

4 Stool / microbiome



- chemically complex and heterogeneous
- homogenize thoroughly
- aliquot consistently
- report wet/dry weight and storage conditions

5 Urine / broth / supernatant



- dilute aqueous matrix
- centrifuge or filter
- creatinine / volume / OD normalization may be needed
- good for secreted or excreted metabolites

Typical preparation logic



Matrix effect
proteins, salts, and lipids can suppress signals

Storage
freeze-thaw cycles change chemistry

Normalization
mass, volume, cell count, or protein

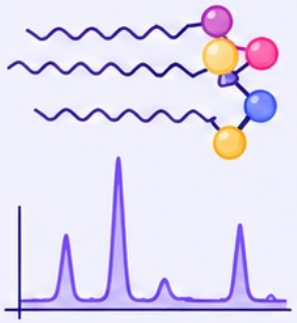
Question first
host metabolism, tissue biochemistry, cell state, or host-microbe chemistry

The matrix is part of the experiment: **choose preparation methods that match both the sample source and the biological question.**

Which analytical platform should I use?

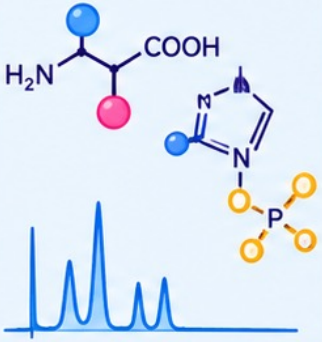
Choose the platform by metabolite chemistry and the biological question, not by popularity alone.

1 RP-LC-MS



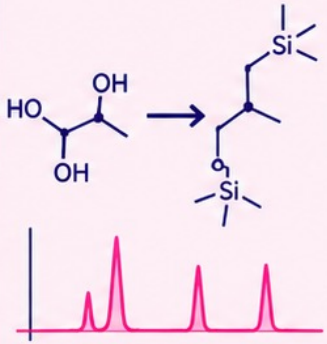
- hydrophobic compounds
- many lipids and bile acids

2 HILIC-LC-MS



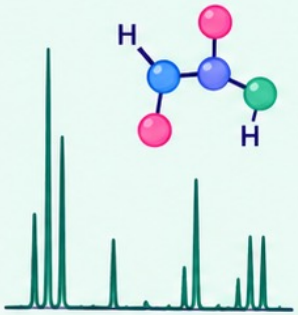
- polar metabolites
- amino acids, nucleotides

3 GC-MS



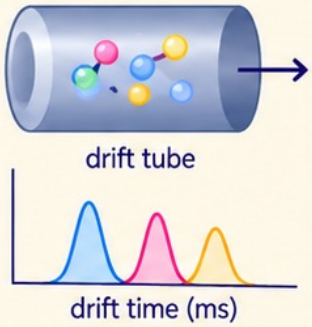
- volatile / derivatized analytes
- organic acids, sugars

4 NMR



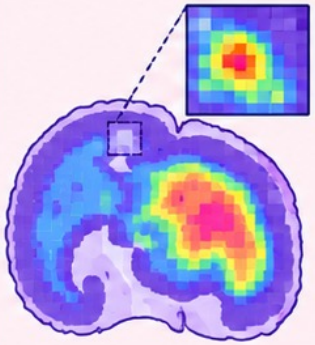
- high reproducibility
- strong quantification for abundant compounds

5 Ion mobility



- extra separation axis
- collision cross section

6 MALDI imaging



- spatial context
- where in the tissue?

Decision hints



Need lipids?



Need polar metabolites?



Need spatial location?



Need strong quantification?

Platform choice follows chemistry, resolution, and the biological question.

How to think about multi-omics integration

Do not start by stacking matrices; start by asking which layer is upstream, functional, and closest to phenotype.

Genomics



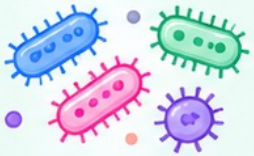
- DNA sequence
- variants (SNP, SV)
- regulatory elements
- genetic background

Transcriptomics



- RNA expression
- splicing
- non-coding RNA
- cell type states

Microbiome



- microbial community
- relative abundance
- species / strain
- functional potential

Proteomics



- protein abundance
- post-translational modifications
- protein complexes
- signaling proteins

Metabolomics



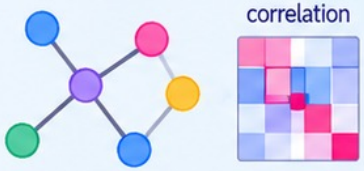
- small molecules
- metabolic pathways
- lipids, amino acids
- host–microbe metabolites

Phenotype



- disease status
- clinical traits
- treatment response
- quantitative outcomes

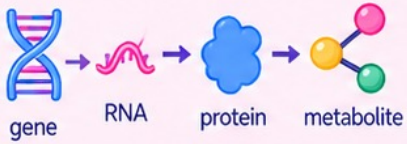
Pairwise associations



- correlate features across layers
- network-based integration
- find cross-omics links

Confounding matters

Pathway mapping



- map features to pathways
- link genes, proteins, metabolites
- use prior biological knowledge

Pathways are incomplete and context-dependent

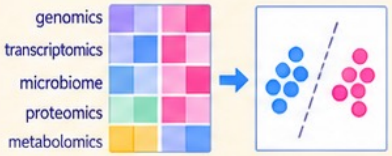
Latent factors (MOFA2)



- learn shared and modality-specific variation
- handle missing data
- identify multi-omics patterns

Shared variation does not automatically imply a pathway

Supervised multi-block (DIABLO)



- find multi-omics signatures discriminating phenotypes
- select correlated features across layers
- predictive and interpretable

Requires careful cross-validation to avoid overfitting

Interpretation rules

- use matched samples across omics layers
- use organism-aware identifiers (host vs microbe)
- account for repeated measures and compositionality
- check feature stability across datasets and methods

The best integration strategy follows the biological logic of the question, not the convenience of the software.

From raw signals to a metabolite matrix

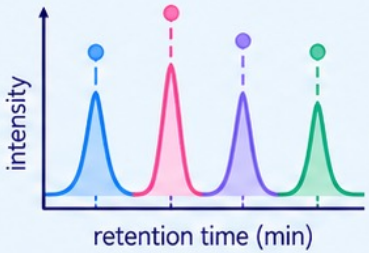
Untargeted metabolomics turns raw files into a feature table through explicit processing decisions.

1 Raw files



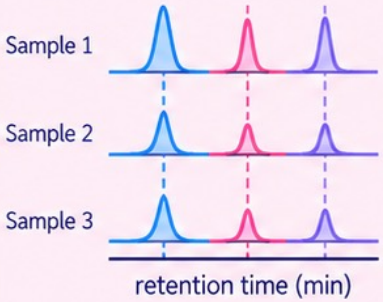
- vendor-specific files (.raw, .wiff, .d, ...)
- convert to mzML
- retain scan metadata (m/z, intensity, RT)

2 Feature detection



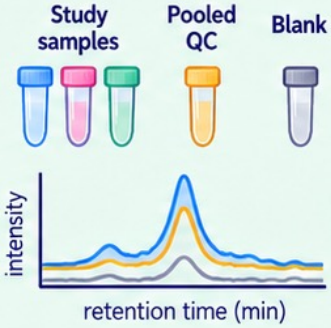
- peak finding
- m/z (mass-to-charge)
- retention time (RT)
- peak area (abundance)

3 Alignment + grouping



- RT correction
- sample-by-feature table
- targeted gap filling (missing values)

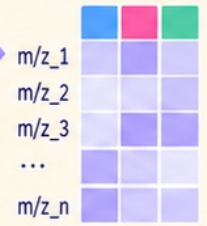
4 QC filtering



- remove blank-dominated features
- exclude unstable QC features
- inspect signal drift

5 Outputs

Feature matrix (quantitative)



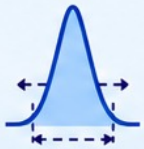
- samples × features (peak areas)

Annotation table



- m/z, RT
- putative annotations
- confidence level

Key reminders



ppm / peak width follow chromatography

- mass accuracy in ppm
- peak width depends on method



adducts + isotopes



- same metabolite can form multiple ions
- look for adducts and isotope patterns



annotation confidence



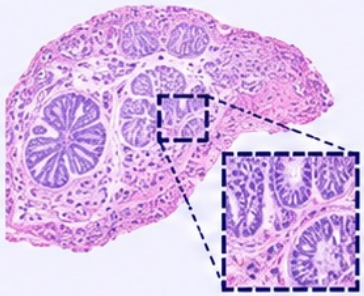
- use MS/MS, libraries, and rules
- report confidence levels

A metabolite table is constructed through processing choices; **reproducibility depends on documented QC and annotation logic.**

Spatial metabolomics asks where chemistry happens

Adding tissue coordinates upgrades the question from what is present to where it is localized.

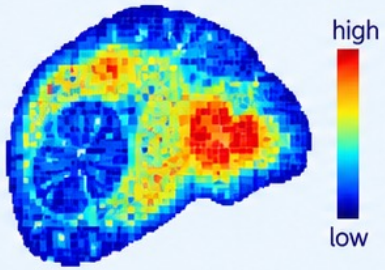
1 Tissue section



- preserve tissue integrity
- define region of interest

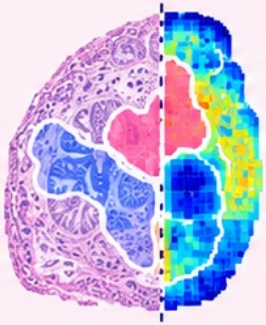
2 Metabolite imaging

MALDI-style pixel map



- ions linked to pixels
- chemical map across the tissue

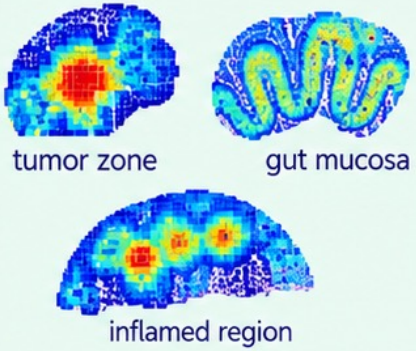
3 Co-registration



- tumor zone
- stroma
- immune region

- histology alignment
- neighborhood context
- compatible resolution

4 What it can reveal

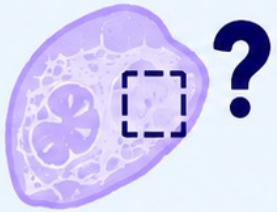


- spatial heterogeneity
- cell-type context
- local metabolic niches

5 Caveats



- matrix effects
- spatial resolution limits
- isomer ambiguity
- local validation still needed



Which region?

- tumor vs stroma
- epithelium vs immune
- normal vs diseased



Which cell type?

- tumor cells
- immune cells
- stromal cells
- epithelial cells



Which local metabolite program?

- lipid metabolism
- amino acid metabolism
- energy metabolism
- signaling molecules

Spatial metabolomics adds location to chemistry, but biological interpretation still depends on careful identification and tissue registration.

AI agents can accelerate analysis, not replace scientific judgment

Use ChatGPT, Gemini, Claude, and coding agents to scaffold analysis, then verify every claim with code, QC, and real sources.

1 Question



- define the study contract
- specify hypothesis
- clarify scope and endpoints

2 Plan



- choose AI tools (ChatGPT, Gemini, Claude, coding agents)
- write effective prompts
- design the workflow

3 Execute



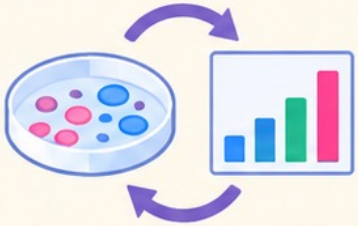
- run R / Python code
- generate tables
- create figures
- document analysis steps

4 Audit



- check sources and citations
- review code for accuracy and reproducibility
- verify statistical methods and assumptions

5 Experiment



- validate results in new data or experiment
- refine analysis or revise hypothesis
- iterate based on findings

What AI helps with



- prompt refinement
- code scaffold
- annotation prioritization
- figure drafting
- explanation

Human checkpoints



- before experiment (review plan and feasibility)
- after analysis (check results and interpretations)
- before final claim (confirm evidence and sources)

Never invent



- annotation
- p-value
- pathway score
- DOI
- or successful tool run

AI should accelerate reproducible work, but final scientific responsibility stays with the researcher.

Functional omics in synthetic biology, biomedicine, and health

Proteomics and metabolomics connect design, mechanism, and translation across modern bioscience.

Synthetic biology/ biomanufacturing



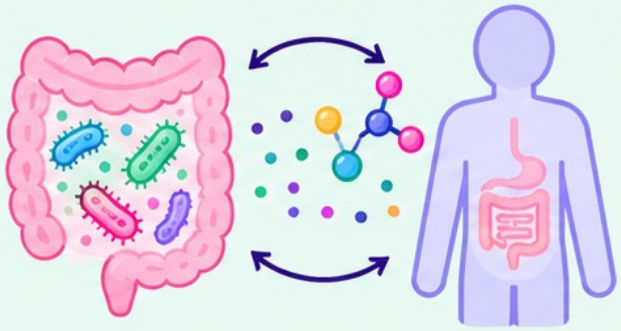
- optimize pathway expression
- quantify enzymes and products
- detect bottlenecks

Biomedicine



- mechanism of action
- biomarkers and disease stratification
- drug response and resistance
- signaling and post-translational modification (PTM) readouts

Life and health science



- host-microbiome interactions
- disease phenotypes and progression
- nutritional or clinical stratification

Question-driven assay choice

- define the biological question
- choose proteomics or metabolomics (or both)
- match to system and sample type

QC before claims

- standardize sampling and processing
- include controls and replicates
- check technical and biological variability

Multi-omics validation

- integrate with genomics, transcriptomics, phenomics
- cross-validate key findings
- link molecular changes to function and phenotype

Functional omics is most powerful when it links molecular measurements to actionable biological questions and testable interventions.

Week 7 synthesis: from measurement to mechanism

Use functional omics to move from data layers to a testable biological claim.

1 Start with the question



- what is the biological object?
- what do you want to learn?
- define scope and hypothesis

2 Choose the assay



- targeted vs. untargeted (e.g. targeted panel, global profiling)
- proteomics vs. metabolomics (choose by question)
- add a mechanism assay when needed (e.g. enzyme activity, binding, flux)

3 Trust the data



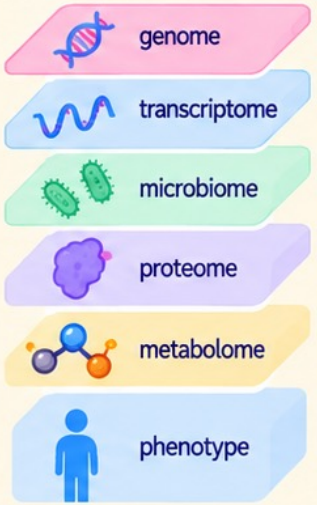
- careful sample preparation
- appropriate controls (blanks, standards)
- quality control (replicates, technical variation)
- calibration and normalization

4 Process + analyze

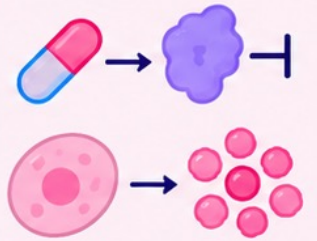


- feature detection and quantification
- statistical testing (differential analysis)
- effect size and biological relevance
- annotation and confidence assessment

5 Integrate layers



6 Validate mechanism



- perturbation (e.g. inhibitor, KO/KD)
- rescue (reverse effect)
- orthogonal assay (independent method)
- measure relevant phenotype or function

Question
biological object and what to learn

→

Assay
targeted / untargeted proteomics / metabolomics (+ mechanism assay)

→

Data
high-quality, well-controlled measurements

→

Analysis
statistics, effect size, annotation confidence

→

Integration
multi-omics evidence across layers

→

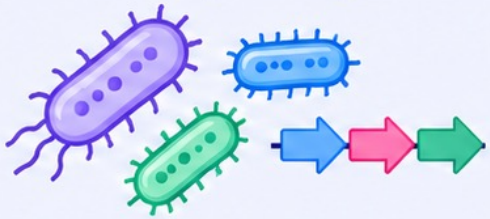
Biological claim
testable mechanism and function

Choose the assay **by the question, trust QC, integrate evidence across layers,** and **validate mechanism** before making strong claims.

Case study: microbe → metabolite → host response

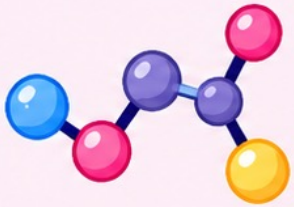
A useful mechanistic model links microbial capacity, chemical mediators, host protein signaling, and phenotype.

1 Microbial gene / pathway



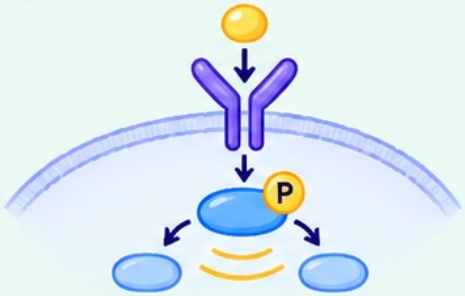
- functional pathway capacity
- candidate producer

2 Chemical signal



- confirmed metabolite
- targeted validation
- source may be microbial or host-microbial

3 Host protein response



- pathway activation
- proteomics or phosphoproteomics readout

4 Phenotype



- metabolism
- inflammation
- growth, or
- disease trait

Causal checks

1 Knock-out / block formation



- remove gene/pathway or inhibit synthesis
- loss of signal and effect

2 Time order



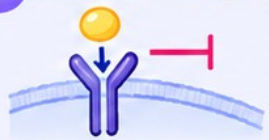
- metabolite changes before host response
- consistent temporal sequence

3 Stable-isotope tracing



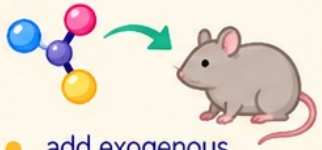
- labelled metabolite from microbe
- detect in host and downstream products

4 Receptor antagonism



- block host receptor or pathway
- abolish or reduce host response

5 Add-back rescue



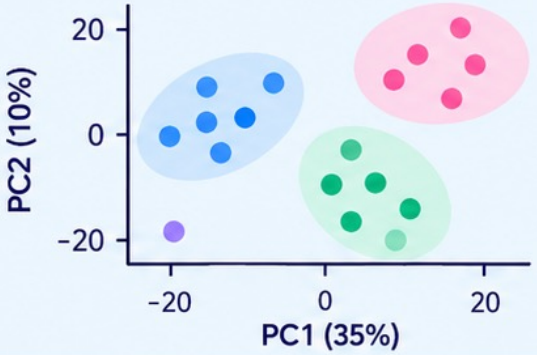
- add exogenous metabolite
- restore host response and phenotype

Association becomes mechanism only after **perturbation**, **temporal logic**, and **rescue** support the proposed mediator.

PCA vs PLS-DA: know what the plot can show

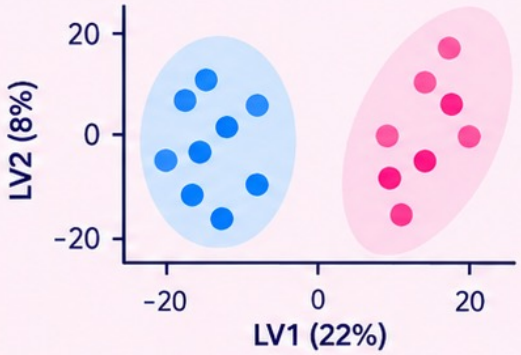
Exploratory and supervised models answer different questions, and neither replaces rigorous validation.

PCA



- unsupervised
- reveals structure or outliers
- projection is not a significance test

PLS-DA



- uses labels (supervised)
- can separate groups
- must be cross-validated
- VIP alone is not validation

Before the model

- **blanks and QCs** — check batch effects and technical noise
- **normalization** — scaling and transformation
- **missingness** — filter or impute consistently
- **covariates** — e.g., age, sex, diet, medication, batch

After the plot

- **permutation testing** — assess if separation is better than random
- **train/test or nested CV** — estimate model performance and avoid overfitting
- **inspect loadings** — check influential variables and potential confounders

Interpretation

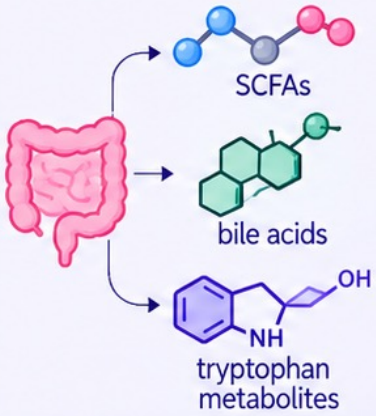
- **separation supports structure** — consistent grouping suggests real differences in the data
- **not mechanism** — a statistical separation does not prove biological causation
- **use effect sizes and adjusted p values** — identify and report important metabolites with appropriate multiple testing control

A separation plot is a starting observation, not a final biological claim.

Targeted metabolomics: from calibration to concentration

A targeted panel becomes analytically valid only when standards, calibration, and QC support the measurement.

1 Define the panel



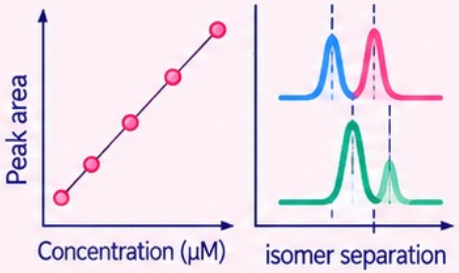
- biological pathway
- expected range
- predefined analytes

2 Standards + internal standards



- matched retention and ionization behavior

3 Calibration + LC-MS/MS

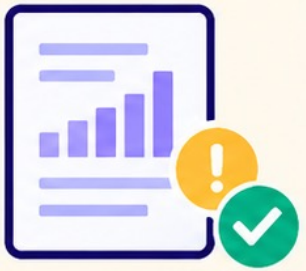


- calibration across physiological range
- critical isomer separation
- transitions for LC-MS/MS

4 Analytical validity



5 Interpretation



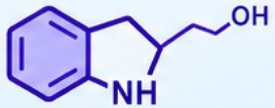
- absolute only within validated range
- independent validation cohort
- QC distributed across batches



SCFAs



bile acids



indole pathway metabolites

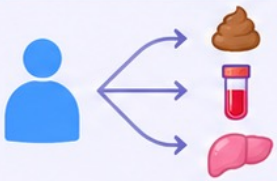
A concentration claim is only as strong as the **calibration, internal standards, and batch-spanning QC** behind it.

Before modeling: build a sample map

Good integration begins with matched samples, metadata, and scale-aware study design.

1 Paired vs unpaired

same subject (paired)



- same biological unit
- links across assays
- more power

different cohorts (unpaired)



- different individuals
- group-level comparison
- harder to integrate

2 Track identifiers



subject ID SUBJ001



sample ID SUBJ001_S1



replicate ID SUBJ001_S1_R1



batch ID BATCH_A01

- consistent naming scheme
- unique and traceable IDs
- link across omics layers

3 Record context



tissue compartment
stool, plasma, tissue, etc.



fasting state
fasted or fed



time point
baseline, week 4, etc.



treatment
drug, placebo, dose, etc.

- capture key covariates
- use standardized terms
- enable downstream stratification

4 Different data scales



microbiome
compositional data (relative abundance)



RNA
counts (raw or normalized)



protein
log2 intensities



metabolite
concentrations (µM, nM, etc.)

- different distributions and units
- require appropriate transforms
- consider zero inflation and compositionality

5 Transparent exclusions

total samples (n = 120)

- missing assays (n = 12)
- repeated measures (n = 8)
- low quality / failed (n = 5)

included samples (n = 95)

- report missing assays
- handle repeated measures
- document collapsed features (e.g., OTUs or metabolites)

Same biological unit? same subject across assays?

Same compartment? tissue, stool, plasma, etc.?

Same time? baseline or longitudinal?

Same batch structure? processed together or separately?

Multi-omics modeling is credible only when the sample map, metadata, and measurement scales are explicit.

Week 7 Summary: Proteomics & Metabolomics

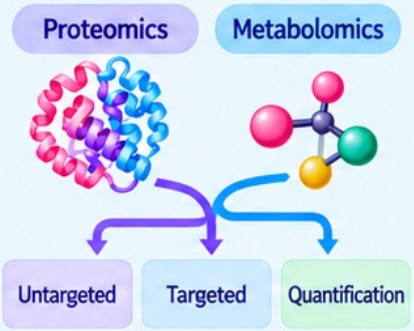
From measurement to mechanism, phenotype, and discovery.

1 Ask the question first



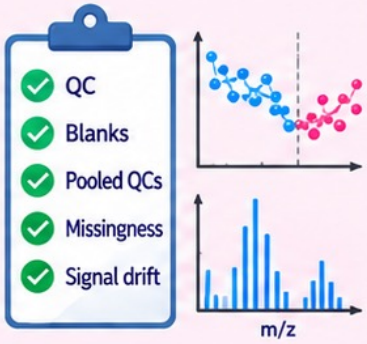
- biology defines the assay
- discovery vs validation
- function vs abundance

2 Choose the right strategy



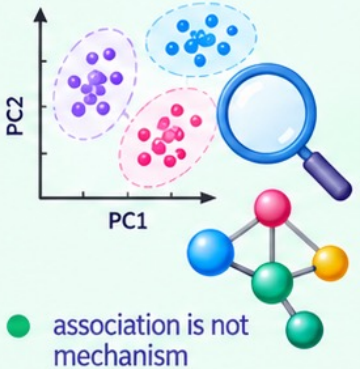
- untargeted = broad discovery
- targeted = precise validation
- acquisition, quantification, targeting are different decisions

3 Trust the data



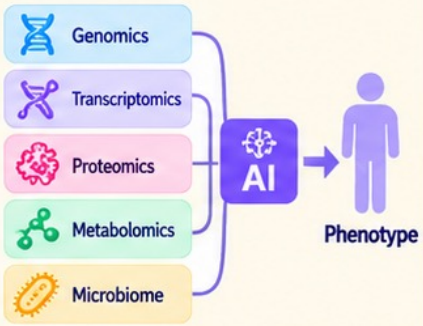
- QC before biology
- inspect drift, missingness, blanks, pooled QCs
- unstable signals should not drive conclusions

4 Interpret carefully



- association is not mechanism
- PCA and PLS-DA are starting points
- proteins, metabolites, PTMs, and interactions answer different questions

5 Integrate and translate



- connect genomics, transcriptomics, microbiome, proteomics, metabolomics
- use multi-omics to support hypotheses
- AI can assist workflows, but scientific judgment stays human

Key learning outcomes

1. Choose assays by the biological question.
2. Separate technical variation from biology.
3. Match statistical claims to the evidence.
4. Build testable multi-omics hypotheses.

The strongest story links good experimental design, reliable measurement, *careful interpretation*, and testable mechanism.

Homework for week 7, **Submission deadline 11, Oct, 23:59 on Sunday Night**

Homework : Progress report of your group project including but not limited to research topics, hypothesis, paper reeading, data collection and analysis strategies.

Important date for your reference

Our oral defense date is **18/25 Nov** and two external professor in the field of bioinformatics will be invited

We will **NOT** have quiz next class.

Do not forget to review the class material.

5 multiple choice questions

1 min of each question for total 5 mins